# Data

## Sample and targets

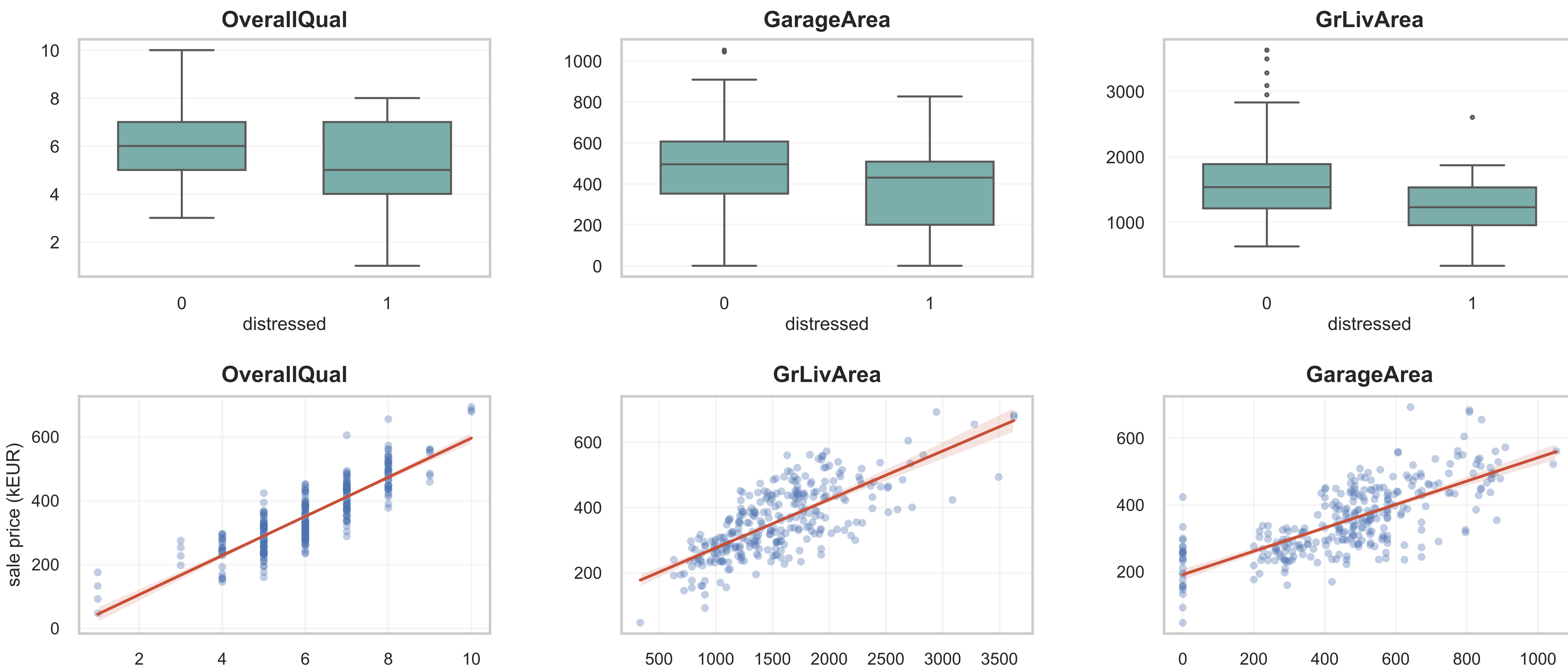
Train / test: 300 / 200
Predictors: 12 numeric variables
Missing values: 0

Sale price mean: 356.3 kEUR
Median: 340.2 kEUR
Range: 47.3-693.4

Distressed:
47 / 300 = 15.7%

## Numerical Pearson correlations

| Predictor | Corr. with price | Corr. with distressed |
|---|---|---|
| GrLivArea | 0.733 | -0.228 |
| LotArea | 0.230 | -0.075 |
| OverallQual | 0.855 | -0.280 |
| OverallCond | -0.119 | -0.255 |
| YearBuilt | 0.654 | -0.147 |
| YearRemodAdd | 0.593 | -0.083 |
| TotalBsmtSF | 0.682 | -0.057 |
| GarageArea | 0.731 | -0.238 |
| BedroomAbvGr | 0.219 | -0.246 |
| Fireplaces | 0.327 | -0.178 |
| WoodDeckSF | 0.287 | -0.055 |
| OpenPorchSF | 0.327 | -0.041 |



# Model Selection & Final Choices

## Libraries

- numpy / pandas: data, arrays, JSON
- sklearn: permitted tree, RF classifier, sigmoid calibration and validation

- imbalanced-learn: fold-local oversampling
- XGBoost 3.2: Tasks 3 and 5

- matplotlib / seaborn: plots and slides
- Tasks 1-2 use self-written CART/RF

## Assumptions & Notes

- Train and test share the same mechanism.
- Rows are independent properties.

- Original year and area values are retained.
- Trees model nonlinearities and interactions.

- Outer validation is untouched by tuning, resampling, early stopping or calibration.
- Importance is predictive, not causal.

## Methods

1. CART tests midpoints and maximizes SSE drop.
2. RF bootstraps rows, samples mtry per node, averages predictions and aggregates MDI.
3. Boosting screens structure and regularization; early stopping uses training data only.
4. Regression selection: validation MSE.
5. Classification selection: stratified log loss; AUC, Brier and prevalence bias support it.
6. Resampling/calibration stay inside train folds.
7. Finalists: six seeds, 30 folds, full-data refit.

## Validation evidence

| Task | Comparison | Selected score | Why selected |
|---|---|---|---|
| 1 CART | 5-fold MSE, depth 2-10 | depth 6 | Required output is depth 3; sklearn agreement exact |
| 2 Scratch RF | OOB screen + 30 CV folds | MSE 1484.85 | Best stable setting: depth 8, leaf 1 |
| 3 XGB reg | early-stop screen + 30 folds | MSE 1362.70 | Lowest regression error; shallow additive trees |
| 4 RF class | stratified probability CV | log loss 0.37046 | Ratio 0.20 resampling beat weighting/calibration |
| 5 XGB class | nested probability CV | log loss 0.38167 | Weight 1.5 + sigmoid gave calibrated probabilities |

## Numerical results

| Model | Validation result | Probability / purity diagnostic | Value |
|---|---|---|---|
| 1 CART | 2311.32 | equiv. max Δ | 2.3e-13 |
| 2 RF reg | 1484.85 | OOB MSE | 1420.81 |
| 3 XGB reg | 1362.70 | top total gain | OverallQual (0.363) |
| 4 RF class | log loss 0.37046 | AUC 0.757 | Brier 0.113 |
| 5 XGB class | log loss 0.38167 | AUC 0.748 | Brier 0.117 |

## Selected model parameters

T1 output depth 3; CV depth 6; leaf 5
SSE drop = purity improvement
T2 B=2000; mtry=4; depth 8; leaf 1
top MDI: OverallQual (0.278)
T3 B=386; eta=.05; depth 1; gamma=.1
top gain: OverallQual (0.363)
T4 B=2000; mtry=6; depth 5; log loss
top MDI: LotArea (0.164)
T5 B=122; eta=.03; depth 3; weight=1.5
top gain: LotArea (0.204)

# Diagnostics and Modeling Insights